

Anton Kuttyrev, PhD

Lecturer in Exploration Geology, Cardiff University Kazakhstan, Astana, Kazakhstan

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Research focus

Petrology and mineralogy of mafic–ultramafic and island-arc systems; melt–rock and fluid–rock interaction; origin of chromitites; PGE mineralogy from hand sample to nanoscale, including alloys and PGE oxides; PGE/PGM cycling and mobility in melts and fluids in ultramafic rocks and layered intrusions; oxygen isotopes as tracers of crustal recycling and early Earth surface environments; critical minerals in the context of the energy transition.

Education

- 2020 **PhD in Geology and Mineral Sciences.**
Defended in 2019 and awarded in 2020 through the Dissertation Council at the V.S. Sobolev Institute of Geology and Mineralogy (SB RAS); research carried out at the Institute of Volcanology and Seismology (RAS), Petropavlovsk-Kamchatsky, Russia.
- 2017 **Specialist degree with Distinction in Applied Geology**, Saint Petersburg Mining University, Russia.
Five-year full-time programme, broadly equivalent to a Master of Science.

Academic appointments

- 2026–present **Lecturer in Exploration Geology**, Cardiff University Kazakhstan, Astana, Kazakhstan.
Teaching, programme leadership, curriculum development and research in economic and exploration geology, including layered intrusions, chromitites, platinum-group elements, critical minerals and isotope geochemistry. Lead the Exploration Geology programme on the Kazakhstan side and coordinate its local delivery; organising the programme’s first fieldwork and developing a new field course from scratch.
- 2025–2026 **Postdoctoral Research Fellow**, School of Geosciences, University of the Witwatersrand, Johannesburg, South Africa.
Critical minerals in layered mafic–ultramafic intrusions and Bushveld chromitites within the DSTI CIMERA industry–public consortium, with a focus on Ni-Cu-PGE systems, chromitites and the petrogenesis of the Critical Zone.
- 2023–2025 **Research Associate**, School of Earth and Environmental Sciences, Cardiff University, Wales, UK.
Critical-mineral and PGE geochemistry in layered intrusions and related volcanic rocks within the Horizon Europe SEMACRET consortium on energy-transition mineral systems, in an interdisciplinary team spanning geology, geophysics and social sciences.
- 2022–2023 **Fulbright Visiting Scholar**, School of Earth Sciences, University of Oregon, USA.
Early Archean climate from triple oxygen and hydrogen isotopes in 3.8 Ga metalavas of the Saglek-Hebron terrane, including development of isotope fractionation models.
- 2020–2022 **Research Associate**, Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia.
PGE mineralization and hydrothermal processes in subduction-related magmatic systems of Kamchatka.

2017–2020	PhD candidate , Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia. Geology and platinum mineralization of concentrically zoned dunite–clinopyroxenite–gabbro complexes of the Tamanvayam and Epilchik groups.
2012–2017	Engineer (0.5 FTE) , Karpinsky Russian Geological Institute, Saint Petersburg, Russia. Regional geology and petrology, including petrography and geochemical data for igneous and metamorphic rocks; work on rock collections of the Academician F.N. Chernyshev Central Geological Research Museum and preparation of a field rock identification guide.

Industry appointments

2018	Geologist (summer fieldwork) , Kirganik Ltd. (Amur Minerals subsidiary), Kamchatka, Russia. Exploration for porphyry Cu and epithermal Au systems: field mapping, alteration and mineralization logging, geochemical sampling.
2016–2017	Geologist (summer field seasons) , Kamp Ltd., Kamchatka, Russia. Junior exploration for PGE mineralization: prospecting, field mapping, heavy-mineral sampling, rock and soil geochemical sampling.
2015	Field technician (summer fieldwork) , Kamchatgeologia, Kamchatka, Russia. Regional geological mapping at 1:200,000 scale: traverses, outcrop description and systematic sampling for petrography and geochemistry.

Teaching experience

2026–present	Cardiff University Kazakhstan . World of Dynamic Environments (EA3504), a Year 1 module on dynamic Earth systems taught through geological and environmental case studies; Earth Materials and fieldwork from this academic year. Designing a new field course from scratch, including its academic structure, field activities and teaching materials.
2025–2026	University of the Witwatersrand . Five-lecture module on magmatic mineral deposits for honours students; demonstrator on the honours field trip (Northern Bushveld).
2023–2025	Cardiff University . Practical sessions for the Mineral Deposits course; guest lectures on mineral deposits; demonstrator on field courses in geological mapping, exploration geology and environmental geoscience (Cyprus, Cornwall, South Wales).
2022–2023	University of Oregon . Demonstrator on petrology field courses; guest lectures in petrology and geohazards.

Formal teaching training

2024	Teaching to Improve the Quality of Student Learning, Cardiff University.
2024	Lecturing Skills, Cardiff University.

Student supervision

2026–present	Co-supervisor (with Prof. Roger Gibson) of the project “Calc-silicate xenoliths in the Lower Zone of the Bushveld Complex and their implications for platinum-group element mineralization” (student: Mensacia Phaswane).
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Additional professional training

2026	PetThermoTools virtual workshop.
2025	Reading the Rocks, Polokwane, South Africa.
2025	Resource Geology Workshop, GeoCongress 2025, Bloemfontein, South Africa.
2024	MDSG workshop “Magmatic Ni-Cu Mineral Systems in Europe”, Edinburgh, UK.

2021	TESCAN electron microscopy course, Saint Petersburg, Russia.
2018	Gold Short Course, Prague, Czech Republic.

Analytical and modelling skills

Hands-on: SEM-EDS, Raman spectroscopy, field-portable XRF, clean-lab chemistry and mass spectrometry for Rb-Sr, Sm-Nd and triple-oxygen isotopes (laser fluorination line), LA-ICP-MS.

Data reduction and interpretation: EBSD, X-ray diffraction, synchrotron micro-XRF/XAS, FIB-TEM, EPMA, atom-probe tomography.

Modelling and data processing: Python and spreadsheet-based workflows; MELTS- and MAGEMin-based thermodynamic modelling; Monte Carlo modelling of triple-oxygen isotope exchange; mass-balance and mixing models.

Scientific service

2024–2025	Coordinator , Minerals and Energy Research Centre, Cardiff University.
2023	Organizing committee member , 14th International Platinum Symposium, Cardiff, UK.
Reviewer	for <i>Ore Geology Reviews</i> , <i>Mineralium Deposita</i> , <i>Nature Communications</i> , <i>Chemical Geology</i> , <i>Journal of Petrology</i> , <i>Australian Journal of Earth Sciences</i> , <i>Contributions to Mineralogy and Petrology</i> , <i>Journal of the Geological Society</i> , <i>Lithos</i> and <i>Marine Geology</i> .

Competitively awarded funding

Amounts converted from foreign currencies to EUR; approximate.

2024	AuScope Access Program (Curtin University), in-kind analytical access: “Silicate–oxide–platinum group minerals intergrowths: implications for unknown aspects of low-T PGE behaviour”. Approx. EUR 10,700.
2022–2023	Fulbright Scholarship (Fulbright Visiting Scholar). Approx. EUR 26,600.
2020–2022	RFBR Project No. 20-35-70008 (project executor): “Formation and alteration of minerals in active volcanic hydrothermal systems (Kamchatka)”. Approx. EUR 114,000.
2020–2021	RFBR Project No. 20-05-00290 (principal investigator): “Distribution of Ni-Fe-Co intermetallic compounds and platinum-group minerals in mafic and ultramafic rocks of various formational types”. Approx. EUR 21,500.
2018–2020	RSCI Project No. 16-17-10145 (project executor): “Melt–fluid evolution, magmatic immiscibility, and metal budgeting in subduction-related magmas”. Approx. EUR 171,000.
2018–2019	RFBR Project No. 18-35-00126 (principal investigator): “Typomorphism and formation conditions of PGE minerals in concentrically zoned ultramafic massifs (Epilchik area, Koryak-Kamchatka belt)”. Approx. EUR 9,700.
Total	Total research funding received: approx. EUR 353,600.

Field work

2025–present	Mapping and sampling of the Critical Zone across the western and eastern Bushveld Complex, South Africa (one month of fieldwork plus regular mine visits).
2023–2025	Fieldwork at the Rhiw layered intrusion, Wales, UK (8 days in total).
2021	Principal investigator and organizer of a helicopter-supported expedition (12 participants) to the Tumrok Range arc picrites, Kamchatka (3 weeks).
2020	Sampling of alluvial awaruite in the Kamchatsky Mys ophiolite, Kamchatka (2 weeks).
2020	Documenting cooling lava flows and caves of the 2012–13 Tolbachik Fissure Eruption (2 weeks); observations of Klyuchevskoy volcano.

2019	Underground and field sampling of mineralized intrusions and related flood basalts, Norilsk (2 weeks).
2018	Documenting lava temperatures in the caves of the 2012–13 Tolbachik Fissure Eruption (2 weeks); visual observations of Shiveluch volcano.
2018	Fieldwork on the Kamchatsky Mys ophiolite (2 weeks).
2018	Prospecting at the Kirganik Cu porphyry and epithermal Au occurrence (1 month).
2016–2017	Prospecting for lode and placer PGE mineralization, Epilchik and Matysken areas, Koryak Highlands (5 months in total).
2015	Geological mapping at 1:200,000 scale, Koryak Highlands (3 months).

Professional societies

2023–present	Mineralogical Society of Great Britain and Ireland.
2021–present	Society of Economic Geologists (SEG).
2017–present	Society of Geology Applied to Mineral Deposits (SGA). 2026–present – Scientific advisor of Central Asian SGA Chapter.
2014–present	Russian Mineralogical Society; 2021–2022 – Head of the Kamchatka Department

Awards

2023	Minerals Young Investigator Award.
2021	Marie Skłodowska-Curie Actions Seal of Excellence.

Public outreach

School visits in the UK and Russia for ages 8 to 16: interactive lectures on volcanoes, minerals and rocks, the societal impact of geology, and career perspectives of a geoscientist.

Publications

Peer-reviewed articles (35)

1. **Kutyrev, A.**, Filimonova, O., Saxey, D.W., Vymazalová, A., Evans, K., Kamenetsky, V.S., Wermeille, D., Fougereuse, D., Reddy, S.M., 2026. Formation of natural IrO₂ through Os-Ir-Ru alloy oxidation. *Chemical Geology*, 123167. <https://doi.org/10.1016/j.chemgeo.2025.123167>
2. Eslami, A., Chardelin, M., O'Driscoll, B., Malvoisin, B., Evans, K., Grieco, G., Bussolesi, M., & **Kutyrev, A.** 2026. Chromitite-mafic dike interfaces as archives of syn-kinematic fluid-rock interaction, chromium mobility and Ca-metasomatism, Nain Ophiolite, Central Iran. *Journal of Petrology*, egag076. <https://doi.org/10.1093/petrology/egag076>
3. Rhoads, E.A., **Kutyrev, A.**, Bindeman, I.N., Lynn, K.J., Trusdell, F.A., Downs, D.T., Edwards, H.R., Cook, G.W., Day, J.M.D., 2025. Rhenium-osmium and oxygen isotope homogeneity during the 2022 Mauna Loa eruption and implications for basaltic magma storage. *Bulletin of Volcanology* 87. <https://doi.org/10.1007/s00445-025-01825-0>
4. Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Barnes, S.J., Stepanov, A.S., Konstantinov, S.E., Baykov, N.I., Kryazhev, S.G., Belousov, I.A., Zhukova, I.A., Scherbakov, V.D., Kalugin, V.M., Ivanov, A.V., Bryanskiy, N.V., Park, J.W., 2025. Erratic behavior of platinum-group elements along an olivine-Cr-spinel cotectic in sulfide-undersaturated basaltic magma: roles of Cr-spinel and alloys. *Scientific Reports* 15, 14366. <https://doi.org/10.1038/s41598-025-97281-z>
5. Zhitova, E.S., Nuzhdaev, A.A., Davydova, V.O., Sheveleva, R.M., Zhegunov, P.S., Kuznetsov, R.A., **Kutyrev, A.V.**, Khokhlova, M.A., Vlasenko, N.S., 2025. Calcareous skarn-like mineral paragenesis from unaltered basalt of the Alaid volcano (Kuril-Kamchatka island arc). *Minerals* 15. <https://doi.org/10.3390/min15030237>

6. **Kutyrev, A.**, Bindeman, I.N., O'Neil, J., Rizo, H., 2024. Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada: Implications for moderately low- $\delta^{18}\text{O}$ Eoarchean ocean. *Chemical Geology* 670. <https://doi.org/10.1016/j.chemgeo.2024.122378>
7. **Kutyrev, A.**, Kamenetsky, V.S., Chayka, I.F., Nekrylov, N.A., Kryuchkova, L., Shilovskikh, V.V., Kontonikas-Charos, A., Krashenninnikov, S.P., Sapegina, A., Perchuk, A.L., 2024. Metamorphic origin of large nuggets of platinum-group metals: evidence from multiphase inclusions in Os-Ir-Ru alloys from the Adamsfield placer, Tasmania. *Journal of Petrology*. <https://doi.org/10.1093/petrology/egae086>
8. Zhegunov, P.S., **Kutyrev, A.V.**, Zhitova, E.S., Moskaleva, S.V., Schweigert, P.E., 2024. First data on the mineralogy of the Evevpena epithermal silver-gold ore occurrence, Kamchatka, Russia. *Journal of Volcanology and Seismology* 18, 149-164. <https://doi.org/10.1134/S0742046324700489>
9. Stepanov, S., Palamarchuk, R., **Kutyrev, A.**, Lepekhina, E., Sharpenok, L., Shagalov, E., Minervina, E., 2024. Nature of perovskite mineralization of silicate-carbonate veins in the margins of Kusinsko-Kopanskaya layered intrusion (South Urals, Russia). *Minerals* 14, 478. <https://doi.org/10.3390/min14050478>
10. **Kutyrev, A.**, McDonald, A., Tamura, N., Cabri, L.J., Tolstykh, N., Pekov, I., 2023. Sidorovite, PtFe_3 , a new mineral from the Snegovaya River placer, Koryak Highlands, Russia. *The Canadian Journal of Mineralogy and Petrology* 61(5), 1021-1030. <https://doi.org/10.3749/2200072>
11. **Kutyrev, A.**, Kamenetsky, V.S., Kontonikas-Charos, A., Savelyev, D.P., Yakich, T.Y., Belousov, I.A., Sandimirova, E.I., Moskaleva, S.V., 2023. Behavior of platinum-group elements during hydrous metamorphism: constraints from awaruite (Ni_3Fe) mineralization. *Lithosphere*, 2023(1). https://doi.org/10.2113/2023/lithosphere_2023_126
12. Chayka, I.F., Izokh, A.E., Kamenetsky, V.S., Sokol, E.V., Lobastov, B.M., Kontonikas-Charos, A., Zelenski, M.E., **Kutyrev, A.V.**, Sluzhenikin, S.F., Zhitova, L.M., Shvedov, G.I., Shevko, A.Y., Gora, M.P., 2023. Origin of chromitites in the Norilsk-1 intrusion (Siberian LIP) triggered by assimilation of argillaceous rocks by Cr-rich basic magma. *Lithos* 454-455. <https://doi.org/10.1016/j.lithos.2023.107254>
13. Eslami, A., Grieco, G., Bussolesi, M., Ichiyama, Y., Lenaz, D., Skogby, H., **Kutyrev, A.V.**, Cavallo, A., Khedr, M.Z., 2023. Co-occurrence of compositionally variable chromitites in the Sabzevar ophiolite, NE Iran. *Lithos* 446-447. <https://doi.org/10.1016/j.lithos.2023.107133>
14. Stepanov, S.Y., Palamarchuk, R.S., **Kutyrev, A.V.**, Shilovskikh, V.V., Petrov, S.V., 2023. Mineralogy and geochemistry of platinum-group elements in the zoned mafic-ultramafic intrusions of the Uralian Platinum belt, Russia. *Journal of Geochemical Exploration* 246. <https://doi.org/10.1016/j.gexplo.2023.107155>
15. Chayka, I.F., Baykov, N.I., Kamenetsky, V.S., **Kutyrev, A.V.**, Pushkarev, E.V., Abersteiner, A., Shcherbakov, V.D., 2023. Volcano-plutonic complex of the Tumrok Range (Eastern Kamchatka): an example of the Ural-Alaskan type intrusion and related volcanic series. *Minerals* 13, 126. <https://doi.org/10.3390/min13010126>
16. Yakubovich, O., **Kutyrev, A.**, Sidorov, E., Travin, A., 2022. ^{190}Pt - ^4He dating of platinum mineralization in Ural-Alaskan-type complexes in the Kamchatka region: evidence for remobilization of platinum-group elements. *Mineralium Deposita*. <https://doi.org/10.1007/s00126-022-01103-5>
17. Bukhanova, D.S., **Kutyrev, A.V.**, Sidorov, E.G., Chubarov, V.M., 2022. First finding of platinum group minerals at the Malmyzh porphyry gold-copper deposit, Khabarovsk Krai, Russia. *Geology of Ore Deposits* 63, 675-681. <https://doi.org/10.1134/s1075701521070035>
18. Pekov, I.V., Zubkova, N.V., Agakhanov, A.A., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Sidorov, E.G., **Kutyrev, A.V.**, Pushcharovsky, D.Y., 2022. New arsenate minerals from the Arsenatnaya fumarole, Tolbachik volcano, Kamchatka, Russia. XIX. Axelite, $\text{Na}_{14}\text{Cu}_7(\text{AsO}_4)_8\text{F}_2\text{Cl}_2$. *Mineralogical Magazine*. <https://doi.org/10.1180/mgm.2022.120>
19. Pekov, I.V., Zubkova, N.V., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Agakhanov, A.A., Turchkova, A.G., Sidorov, E.G., **Kutyrev, A.V.**, Blatov, V.A., Pushcharovsky, D.Y., 2022. Nishanbaevite, $\text{KAl}_2\text{O}(\text{AsO}_4)(\text{SO}_4)$, a new As/S-ordered arsenate-sulfate mineral of fumarolic origin. *Mineralogy and Petrology*. <https://doi.org/10.1007/s00710-022-00803-0>

20. Cui, M.-M., Su, B.-X., Wang, J., Wu, Y., **Kutyrev, A.**, 2022. Early Paleozoic subduction fingerprints of the Paleo-Asian ocean in easternmost Central Asian Orogenic Belt (CAOB): Identification of the oldest Alaskan-type complex in the CAOB. *GSA Bulletin*. <https://doi.org/10.1130/B36307.1>
21. Nekrylov, N., Kamenetsky, V.S., Savelyev, D.P., Gorbach, N.V., Kontonikas-Charos, A., Palesskii, S.V., Shcherbakov, V.D., **Kutyrev, A.V.**, Savelyeva, O.L., Korneeva, A.A., Kozmenko, O.A., Zelenski, M.E., 2022. Platinum-group elements in Late Quaternary high-Mg basalts of eastern Kamchatka: Evidence for minor cryptic sulfide fractionation in primitive arc magmas. *Lithos*. <https://doi.org/10.1016/j.lithos.2022.106608>
22. **Kutyrev, A.V.**, Kamenetsky, V.S., Park, J.-W., Maas, R., Demonterova, E.I., Antsiferova, T.N., Ivanov, A.V., Hwang, J., Abersteiner, A., Ozerov, A.Y., 2021. Primitive high-K intraoceanic arc magmas of Eastern Kamchatka: Implications for Paleo-Pacific tectonics and magmatism in the Cretaceous. *Earth-Science Reviews* 220, 1-22. <https://doi.org/10.1016/j.earscirev.2021.103703>
23. **Kutyrev, A.**, Zelenski, M., Nekrylov, N., Savelyev, D., Kontonikas-Charos, A., Kamenetsky, V.S., 2021. Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano. *Frontiers in Earth Science* 9. <https://doi.org/10.3389/feart.2021.791465>
24. **Kutyrev, A.V.**, Sidorov, E.G., Kamenetsky, V.S., Chubarov, V.M., Chayka, I.F., Abersteiner, A., 2021. Platinum mineralization and geochemistry of the Matysken zoned Ural-Alaskan type complex and related placer (Far East Russia). *Ore Geology Reviews* 130, 1-17. <https://doi.org/10.1016/j.oregeorev.2020.103947>
25. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Agakhanov, A.A., Sandimirova, E.I., Vymazalová, A., Chubarov, V.M., Zolotarev, A.A., 2021. Kufahrite, PtPb, a new mineral from Ledyanoy Creek placer, Galmoenan ultramafic complex, Koryak Highlands, Russia. *Mineralogical Magazine* 85, 254-261. <https://doi.org/10.1180/mgm.2021.18>
26. Stepanov, S.Y., **Kutyrev, A.V.**, Lepekhina, E.N., Sharpenok, L.N., Antonov, A.V., Kutyreva, M.E., 2021. Age of the dike complex in the dunite "core" of the Kamenushinsky clinopyroxenite-dunite massif, Ural Platinum belt, Middle Urals. *Geochemistry International* 59, 559-576. <https://doi.org/10.1134/s0016702921060094>
27. Sidorov, E., **Kutyrev, A.**, Chubarov, V., Zhitova, E., 2021. Platinum mineralization of the Epilchik Ural-Alaskan type zoned complex (Far East Russia). *Mineralium Deposita* 56, 143-160. <https://doi.org/10.1007/s00126-020-01008-1>
28. Ismagilova, R.M., Zhitova, E.S., Krivovichev, S.V., Sergeeva, A.V., Nuzhdaev, A.A., Anikin, L.P., Krzhizhanovskaya, M.G., Nazarova, M.A., Kupchenko, A.N., Zolotarev, A.A., **Kutyrev, A.V.**, Bukhanova, D.S., Kuznetsov, R.A., Khanin, D.A., 2021. Phase evolution from volborthite, $\text{Cu}_3(\text{V}_2\text{O}_7)(\text{OH})_2 \cdot 2\text{H}_2\text{O}$, upon heat treatment. *Minerals* 11, 1312. <https://doi.org/10.3390/min11121312>
29. Saveliev, D.E., Shilovskikh, V.V., Sergeev, S.N., **Kutyrev, A.V.**, 2021. Chromian spinel neomineralisations and the microstructure of plastically deformed ophiolitic peridotites (Kraka massifs, Southern Urals, Russia). *Mineralogy and Petrology* 115, 411-430. <https://doi.org/10.1007/s00710-021-00748-w>
30. **Kutyrev, A.V.**, Kamenetsky, V.S., Sidorov, E.G., Abersteiner, A., Chubarov, V.M., 2020. Silicate inclusions in isoferroplatinum: Constraints on the origin of platinum mineralization in podiform chromitites. *Ore Geology Reviews* 119, 1-13. <https://doi.org/10.1016/j.oregeorev.2020.103367>
31. Chayka, I.F., Sobolev, A.V., Izokh, A.E., Batanova, V.G., Krashennnikov, S.P., Chervyakovskaya, M.V., Kontonikas-Charos, A., **Kutyrev, A.**, Lobastov, B., Chervyakovskiy, V., 2020. Fingerprints of Kamafugite-Like Magmas in Mesozoic Lamproites of the Aldan Shield: Evidence from Olivine and Olivine-Hosted Inclusions. *Minerals* 10, 1-30. <https://doi.org/10.3390/min10040337>
32. Stepanov, S.Y., Shagalov, E.S., Palamarchuk, R.S., **Kutyrev, A.V.**, Sharpenok, L.N., Nabiullin, F.M., Troshkina, A.N., 2020. Mineralogical and geochemical evidence for granite metasomatism in dikes in the north of the Berezovskoe Ore field (Middle Urals). *Russian Geology and Geophysics* 61, 809-827. <https://doi.org/10.15372/RGG2019176>
33. **Kutyrev, A.V.**, Zhirnova, T.S., 2019. Concentrically-zoned massifs of the Tamanvayam group (Koryak-Kamchatka Platiniferous Belt): structure, age, petrological and geochemical aspects. *Russian Journal of Pacific Geology* 13, 350-363. <https://doi.org/10.1134/S1819714019040031>

34. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Chubarov, V.M., Khanin, D.A., 2019. Origin of platinum-group mineral assemblages from placers in rivers draining from the Ural-Alaskan type Itchayvayamsky ultramafics, Far East Russia. *Canadian Mineralogist* 57, 91-104. <https://doi.org/10.3749/canmin.1800040>
35. **Kutyrev, A.V.**, Sidorov, E.G., Antonov, A.V., Chubarov, V.M., 2018. Platinum-group mineral assemblage of the Prizhimny Creek (Koryak Highland). *Russian Geology and Geophysics* 59, 935-944. <https://doi.org/10.1016/j.rgg.2018.07.014>

Under review

1. **Kutyrev, A.**, Maier, W., Karinen, T., Barnes, S.-J., Hayes, B., Savard, D., Virtanen, V., Booth, C., Muir, D. Petrogenesis of the Akanvaara layered intrusion, Finnish Lapland. *Journal of Petrology*, under review.
2. **Kutyrev, A.**, Kamenetsky, V.S., Chayka, I.F., Yudovskaya, M.A., Eslami, A., Kislov, E.V., Pushkarev, E.V., Zhitova, L.M. What do Cr-spinel-hosted multiphase inclusions in chromitites record? *Journal of Petrology*, submitted.
3. Chayka, I., Hu, W.-J., **Kutyrev, A.**, Petrenko, D., Zhitova, L., Shevko, A., Gora, M., Kalugin, V., Shvedov, G., Izokh, A., Yudovskaya, M. Fluid-mediated PGE enrichment of magmatic low-sulfide ores as modelled for the Norilsk deposits. *Geology*, under review.
4. Baykov, N.I., Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Nekrylov, N.A., Shcherbakov, V.D., Zhukova, I.A., Plechov, P.Yu., Abramova, V.D., Stepanov, A.S. Melt-spinel interface effects control Pt fractionation in basaltic melt. *Geology*, submitted.

Datasets and code

1. **Kutyrev, A.V.**, Maier, W.D., Karinen, T., Barnes, S.J., Savard, D., 2025. Whole-rock geochemical dataset of the Akanvaara layered intrusion, Lapland, Finland. Dataset deposited with the Interdisciplinary Earth Data Alliance (IEDA). <https://doi.org/10.60520/IEDA/113569>
2. Filimonova, O., **Kutyrev, A.**, Wermeille, D., 2025. Iridium L₃-edge XAS dataset of natural Ir-O phase, comprising IrO₂ [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.15650871>
3. Chayka, I.F., Kamenetsky, V.S., **Kutyrev, A.V.**, Baykov, N.I., Barnes, S., Savard, D., Alexei, I.V., 2025. Major, trace, and platinum-group elements in the Late Cretaceous arc volcano-plutonic complex of the Tumrok Range, eastern Kamchatka, Version 1.0. Interdisciplinary Earth Data Alliance (IEDA). <https://doi.org/10.60520/IEDA/113578>
4. **Kutyrev, A.**, Bindeman, I., O'Neil, J., Rizo, H., 2024. Code for 1- and 2-stage oxygen fractionation and metamorphic dehydration modeling from "Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada". <https://doi.org/10.5281/zenodo.12795578>

Conference presentations

1. IMSG 2026, Barberton, South Africa. Oral presentation: "Mottled anorthosites and discordant ultramafic pipes in the MG4-UG1 interval of the Upper Critical Zone, Bushveld Complex".
2. Geocongress 2025, Bloemfontein, South Africa. Oral presentation: "Petrogenesis of the Akanvaara layered intrusion".
3. MDSG 2024, Edinburgh. Poster: "Assessing mass-balance of layered intrusions: the Akanvaara intrusion case study in Lapland, Finland".
4. Goldschmidt 2023, Lyon, France. Remote oral talk: "Early Earth's hydrosphere: insights from the hydrothermally altered submarine lavas of the Eoarchean Saglek-Hebron Complex, Canada".
5. 14th International Platinum Symposium 2023, Cardiff, UK. In-person oral presentation: "Origin of native osmium in metamorphosed peridotites: insights from multiphase inclusions (Adamsfield placer, Tasmania)".
6. Virtual Early Career Platinum Symposium 2022. Oral presentation: "Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano, Kamchatka".
7. EGU General Assembly 2021. Oral presentation: "Awaruite (Ni₃Fe) as a platinum-group elements concentrator: Preliminary data".

8. SGA 2019 Conference, Glasgow, United Kingdom. In-person oral presentation: “Platinum mineralization in Ural-Alaskan complexes of Koryak Highlands (Far East Russia)”.